



### SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians"
- "Authenticate librarians"
- "Validate user sessions"
- "Manage credentials securely"

dependencies:

- "Audit Service"

suggested\_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books in the collection"
- "Delete registered books"
- "List registered books"
- "Consult books by identifier"
- "Consult books by title"
- "List books available for loan"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested\_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans"
- "Delete registered readers"
- "List registered readers"
- "Consult readers by identifier"
- "Consult readers by name"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested\_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans"
- "Record loan dates"
- "Ensure single active loan per book"
- "Remove loan links"
- "Validate reader and book data for loans"

dependencies:

- "Book Catalog Service"
- "Reader Management Service"
- "Audit Service"

suggested\_endpoints:

- "/api/loan-management-service"

- name: "Environment Management Service"

responsibilities:

- "Clear data and base structure"
- "Recreate base structure"
- "Coordinate functional areas to reset environment"

dependencies:

- "Authentication Service"
- "Book Catalog Service"
- "Reader Management Service"
- "Loan Management Service"
- "Audit Service"

suggested\_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Sum two integers"
- "Calculate Fibonacci sequence"
- "Calculate Ackermann function"
- "Provide service catalog for testing endpoints"

dependencies:

- ""

suggested\_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Record relevant operations from system flows"
- "Maintain decoupled audit and behavior tracking"

dependencies:

- ""

suggested\_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"  
to: "Audit Service"
- from: "Book Catalog Service"  
to: "Loan Management Service"
- from: "Book Catalog Service"  
to: "Audit Service"
- from: "Reader Management Service"  
to: "Loan Management Service"
- from: "Reader Management Service"  
to: "Audit Service"
- from: "Loan Management Service"  
to: "Book Catalog Service"
- from: "Loan Management Service"  
to: "Reader Management Service"
- from: "Loan Management Service"  
to: "Audit Service"
- from: "Environment Management Service"  
to: "Authentication Service"
- from: "Environment Management Service"  
to: "Book Catalog Service"
- from: "Environment Management Service"  
to: "Reader Management Service"
- from: "Environment Management Service"  
to: "Loan Management Service"
- from: "Environment Management Service"  
to: "Audit Service"

## SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Create and manage usage sessions"
- "Validate active sessions"
- "Invalidate sessions on logout and expiration"

dependencies:

- ""

suggested\_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Search one-way and round-trip flights"
- "Present flight options with schedules and reference fares"
- "Manage registered airports and routes"

dependencies:

- ""

suggested\_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create one-way and round-trip bookings"
- "Generate booking identifiers and record booking dates"
- "Consult and list customer bookings"
- "Cancel bookings and update active status"

dependencies:

- "Authentication Service"
- "Flight Search Service"

suggested\_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Consult customer profile and contact data"
- "Update contact and address information"
- "Validate profile modifications with current password"

dependencies:

- "Authentication Service"

suggested\_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Present loyalty program status"
- "Provide annual and total accumulated miles"

dependencies:

- "Authentication Service"

suggested\_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Load and reload application database data"
- "Generate sample operational data"
- "Provide summarized operational indicators and metrics"

dependencies:

- "Authentication Service"
- "Flight Search Service"

- "Booking Service"
- "Customer Profile Service"

suggested\_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"  
to: "Authentication Service"
- from: "Booking Service"  
to: "Flight Search Service"
- from: "Customer Profile Service"  
to: "Authentication Service"
- from: "Loyalty Service"  
to: "Authentication Service"
- from: "Environment Administration Service"  
to: "Authentication Service"
- from: "Environment Administration Service"  
to: "Flight Search Service"
- from: "Environment Administration Service"  
to: "Booking Service"
- from: "Environment Administration Service"  
to: "Customer Profile Service"

## SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Cargo Management Service"

responsibilities:

- "Register new cargo with origin destination and arrival deadline"
- "Assign unique tracking identifier to cargo"
- "Enforce fixed origin and distinct destination validation"
- "Manage cargo lifecycle states"
- "Update cargo destination and request route reevaluation or replanning"

dependencies:

- "Route Planning Service"

suggested\_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Calculate and present candidate routes matching cargo criteria and deadline"
- "Validate and assign chosen route to cargo"
- "Maintain route legs voyages load points unload points and schedules"
- "Reevaluate and replan routes for off-route cargos or destination changes"
- "Validate voyage existence for logistics events"

dependencies:

- "Cargo Management Service"

suggested\_endpoints:

- "/api/route-planning-service"

- name: "Logistics Event Registration Service"

responsibilities:

- "Receive and record logistics events including reception load unload customs and pickup"
- "Validate event data against existing cargo and voyage"
- "Reject and record invalid event registration attempts"
- "Detect route adherence and flag off-route execution"
- "Trigger recalculation of transport status last known location and cargo completion"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Cargo Tracking Service"

suggested\_endpoints:

- "/api/logistics-event-registration-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Consolidate cargo operational status and event history"
- "Track last known location and estimated arrival time"
- "Distinguish expected from unexpected events and highlight off-route deviations"
- "Provide geographic visualization of cargo route and current progress"
- "Provide consolidated cargo tracking data for queries and monitoring"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Logistics Event Registration Service"

suggested\_endpoints:

- "/api/cargo-tracking-service"

- name: "Integration Service"

responsibilities:

- "Ingest logistics events from multiple operational channels including files and interfaces"
- "Process periodic batch input files and isolate invalid records"
- "Expose standardized cargo tracking and status views to external systems and monitoring channels"

dependencies:

- "Logistics Event Registration Service"
- "Cargo Tracking Service"

suggested\_endpoints:

- "/api/integration-service"

interactions:

- from: "Cargo Management Service"
- to: "Route Planning Service"

- from: "Route Planning Service"  
to: "Cargo Management Service"
- from: "Logistics Event Registration Service"  
to: "Cargo Management Service"
- from: "Logistics Event Registration Service"  
to: "Route Planning Service"
- from: "Logistics Event Registration Service"  
to: "Cargo Tracking Service"
- from: "Cargo Tracking Service"  
to: "Cargo Management Service"
- from: "Cargo Tracking Service"  
to: "Route Planning Service"
- from: "Cargo Tracking Service"  
to: "Logistics Event Registration Service"
- from: "Integration Service"  
to: "Logistics Event Registration Service"
- from: "Integration Service"  
to: "Cargo Tracking Service"

## SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate investor credentials using username and password"
- "Start and manage active investor sessions"
- "End investor sessions through logout"
- "Prevent unauthenticated access to restricted resources"

dependencies:

- "Account Service"

suggested\_endpoints:

- "/api/authentication-service"

- name: "Account Service"

responsibilities:

- "Register new investors"
- "Validate identical password and password confirmation during registration and profile update"
- "Prevent registration with existing identification"
- "Maintain investor account records including balances and access statistics"
- "Consult investor registration and financial data"
- "Update investor profile"
- "Process balance debits and credits for trading operations"

dependencies:

- "Authentication Service"

suggested\_endpoints:

- "/api/account-service"

- name: "Market Data Service"

responsibilities:

- "Query asset quotes individually and simultaneously"
- "Provide quote details including volume and price range and current price and variation"
- "Provide consolidated market summary with performance index and top gainers and losers"
- "Handle invalid asset codes safely"
- "Update traded volume and market indicators after buy and sell operations"

dependencies:

- "Trading Service"
- "Simulation Administration Service"

suggested\_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Create trading orders for buying and selling assets"
- "Apply fixed operation fees to buy and sell orders"
- "Process orders immediately or later based on simulation configuration"
- "Track order lifecycle across open processing completed and cancelled statuses"
- "Prevent duplicate completion of orders"
- "Cancel sell orders when informed position is unavailable"
- "Provide order history and summarized recent orders"
- "Alert investor about completed orders"

dependencies:

- "Account Service"
- "Portfolio Service"
- "Market Data Service"
- "Simulation Administration Service"

suggested\_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor portfolio positions"
- "Register new positions on buy order completion"
- "Remove positions on sell order completion"
- "Consult investor portfolio and position details"
- "Allow selling positions directly from portfolio"
- "Calculate total invested amount and market value and consolidated gain or loss"
- "Provide consolidated financial position summary including cash balance and market value and accumulated

dependencies:

- "Account Service"
- "Trading Service"



- "Market Data Service"

suggested\_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters"
- "Manage order processing mode and simulated limits and refresh intervals"
- "Enable or disable completed order alerts"
- "Prepare and rebuild simulation database"
- "Reset simulation operational data"
- "Provide summarized execution statistics and monitoring indicators"

dependencies:

- "Account Service"
- "Trading Service"
- "Portfolio Service"
- "Market Data Service"

suggested\_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"  
to: "Account Service"
- from: "Account Service"  
to: "Authentication Service"
- from: "Market Data Service"  
to: "Trading Service"
- from: "Market Data Service"  
to: "Simulation Administration Service"
- from: "Trading Service"  
to: "Account Service"
- from: "Trading Service"  
to: "Portfolio Service"
- from: "Trading Service"  
to: "Market Data Service"
- from: "Trading Service"  
to: "Simulation Administration Service"
- from: "Portfolio Service"  
to: "Account Service"
- from: "Portfolio Service"  
to: "Trading Service"
- from: "Portfolio Service"  
to: "Market Data Service"
- from: "Simulation Administration Service"  
to: "Account Service"

- from: "Simulation Administration Service"  
to: "Trading Service"  
- from: "Simulation Administration Service"  
to: "Portfolio Service"  
- from: "Simulation Administration Service"  
to: "Market Data Service"

## SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users preventing duplicate usernames"
- "Authenticate user credentials"
- "Manage regular user and administrator profiles"
- "Control user sessions and logout"

dependencies:

- ""

suggested\_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "List movies with pagination and total count"
- "Present movie details and full cast"
- "Manage categories and filter movies by category"
- "Register new movies ensuring uniqueness by title"
- "Delete movies individually and in batch"

dependencies:

- "Authentication Service"

suggested\_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Upload video files"
- "Manage video files between local repository and delivery area"
- "Deliver video content for playback"
- "Provide download links for movie files"
- "Verify video file availability for registered titles"

dependencies:

- "Authentication Service"
- "Catalog Service"

suggested\_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"  
to: "Authentication Service"
- from: "Media Service"  
to: "Authentication Service"
- from: "Media Service"  
to: "Catalog Service"

## SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Organize products by animal categories"
- "Provide category navigation and display product lists"
- "Display available items and item details"
- "Search products by keyword and validate search input"

dependencies:

- "Inventory Service"

suggested\_endpoints:

- "/api/catalog-service"

- name: "Customer Account Service"

responsibilities:

- "Register new customer profiles and addresses"
- "Manage customer profile data and preferences"
- "Configure favorite category and personalized product recommendations and banners"

dependencies:

- "Authentication Service"

suggested\_endpoints:

- "/api/customer-account-service"

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Manage customer login sessions and logout"
- "Provide access control for protected features"

dependencies:

- "Customer Account Service"

suggested\_endpoints:

- "/api/authentication-service"

- name: "Shopping Cart Service"

responsibilities:

- "Manage temporary item selections in shopping cart"
- "Consolidate items and update or remove quantities"

- "Calculate shopping cart subtotals"
- "Display item stock status at addition"

dependencies:

- "Catalog Service"
- "Inventory Service"

suggested\_endpoints:

- "/api/shopping-cart-service"

- name: "Inventory Service"

responsibilities:

- "Track item stock availability and backorder status"
- "Provide updated stock information"
- "Register stock decrements upon order completion"

dependencies:

- "Catalog Service"

suggested\_endpoints:

- "/api/inventory-service"

- name: "Order Service"

responsibilities:

- "Manage checkout process and order lifecycle"
- "Pre-fill billing and delivery data from customer registration"
- "Compose orders from shopping cart items"
- "Collect and adjust payment and shipping address details"
- "Calculate order totals and record completed orders"
- "Clear shopping cart after purchase completion"
- "Manage order history and provide order details consultation"

dependencies:

- "Authentication Service"
- "Customer Account Service"
- "Shopping Cart Service"
- "Inventory Service"

suggested\_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"  
to: "Inventory Service"
- from: "Customer Account Service"  
to: "Authentication Service"
- from: "Authentication Service"  
to: "Customer Account Service"
- from: "Shopping Cart Service"  
to: "Catalog Service"
- from: "Shopping Cart Service"  
to: "Inventory Service"

- from: "Inventory Service"  
to: "Catalog Service"  
- from: "Order Service"  
to: "Authentication Service"  
- from: "Order Service"  
to: "Customer Account Service"  
- from: "Order Service"  
to: "Shopping Cart Service"  
- from: "Order Service"  
to: "Inventory Service"

## SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owners"
- "Update owner data"
- "Search owners by last name prefix or broad query"
- "Display detailed owner profile and paginated search results"

dependencies:

- ""

suggested\_endpoints:

- "/api/owner-service"

- name: "Animal Service"

responsibilities:

- "Register animals linked to owners"
- "Update animal data"
- "Provide predefined list of animal types"
- "Validate animal business rules including unique names per owner and birth dates"

dependencies:

- "Owner Service"

suggested\_endpoints:

- "/api/animal-service"

- name: "Visit Service"

responsibilities:

- "Register visits for animals"
- "Maintain animal visit history"
- "Ensure consistency during veterinarian assignment"

dependencies:

- "Animal Service"
- "Veterinarian Service"

suggested\_endpoints:

- "/api/visit-service"

- name: "Veterinarian Service"

responsibilities:

- "Manage veterinarian profiles and specialties"
- "List veterinarians in paginated and structured formats"
- "Calculate veterinarian workload and next expected visits"
- "Automatically assign visits to veterinarians prioritizing lowest workload"

dependencies:

- "Visit Service"

suggested\_endpoints:

- "/api/veterinarian-service"

interactions:

- from: "Animal Service"  
to: "Owner Service"
- from: "Visit Service"  
to: "Animal Service"
- from: "Visit Service"  
to: "Veterinarian Service"
- from: "Veterinarian Service"  
to: "Visit Service"

## SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration and Configuration Service"

responsibilities:

- "Maintain support records including departments"

dependencies:

- "contract types"

suggested\_endpoints:

- "/api/administration-and-configuration-service"

- name: "Identity and Access Service"

responsibilities:

- "Register and maintain users with personal"

dependencies:

- "contractual"

suggested\_endpoints:

- "/api/identity-and-access-service"

- name: "Workforce Management Service"

responsibilities:

- "Maintain work agreements and vacation requests"
- "Control vacation request states and calculate balances"
- "Trigger notifications for vacation approvals and contract expiration"

dependencies:

- "Administration and Configuration Service"
- "Identity and Access Service"
- "Notification Service"

suggested\_endpoints:

- "/api/workforce-management-service"

- name: "Project Tracking Service"

responsibilities:

- "Maintain projects linked to clients"
- "Register internal and external activities"
- "Classify activities and calculate worked hours"
- "Support evidence policies"

dependencies:

- "Identity and Access Service"
- "Workforce Management Service"
- "File Storage Service"
- "Notification Service"

suggested\_endpoints:

- "/api/project-tracking-service"

- name: "CRM and Proposals Service"

responsibilities:

- "Maintain organizations"

dependencies:

- "contacts"

suggested\_endpoints:

- "/api/crm-and-proposals-service"

- name: "Financial Management Service"

responsibilities:

- "Maintain financial accounts and entries"
- "Calculate and monitor financial indicators"
- "Import billable hours and costs"

dependencies:

- "Billing Service"
- "Project Tracking Service"

suggested\_endpoints:

- "/api/financial-management-service"

- name: "Billing Service"

responsibilities:

- "Detail invoices and calculate totals"

dependencies:

- "taxes"

suggested\_endpoints:

- "/api/billing-service"

- name: "Content and Document Management Service"

responsibilities:

- "Organize documents by categories and versions"
- "Associate documents with users and organizational areas"

dependencies:

- "Identity and Access Service"
- "File Storage Service"

suggested\_endpoints:

- "/api/content-and-document-management-service"

- name: "Commissioning Service"

responsibilities:

- "Register and manage orders and commissions"

dependencies:

- "Project Tracking Service"
- "Workforce Management Service"
- "File Storage Service"
- "Notification Service"

suggested\_endpoints:

- "/api/commissioning-service"

- name: "Reporting Service"

responsibilities:

- "Consolidate analytical and operational data"
- "Export reports across business domains"

dependencies:

- "Administration and Configuration Service"
- "Identity and Access Service"
- "Workforce Management Service"
- "Project Tracking Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Content and Document Management Service"
- "Commissioning Service"

suggested\_endpoints:

- "/api/reporting-service"

- name: "Notification Service"

responsibilities:

- "Send email notifications"
- "Maintain notification templates and delivery mechanisms"

dependencies:

- "Administration and Configuration Service"

suggested\_endpoints:

- "/api/notification-service"



- name: "File Storage Service"

responsibilities:

- "Store files and digital evidence"
- "Retrieve stored files"

dependencies:

- "Administration and Configuration Service"

suggested\_endpoints:

- "/api/file-storage-service"

interactions:

- from: "Administration and Configuration Service"  
to: "contract types"
- from: "Identity and Access Service"  
to: "contractual"
- from: "Workforce Management Service"  
to: "Administration and Configuration Service"
- from: "Workforce Management Service"  
to: "Identity and Access Service"
- from: "Workforce Management Service"  
to: "Notification Service"
- from: "Project Tracking Service"  
to: "Identity and Access Service"
- from: "Project Tracking Service"  
to: "Workforce Management Service"
- from: "Project Tracking Service"  
to: "File Storage Service"
- from: "Project Tracking Service"  
to: "Notification Service"
- from: "CRM and Proposals Service"  
to: "contacts"
- from: "Financial Management Service"  
to: "Billing Service"
- from: "Financial Management Service"  
to: "Project Tracking Service"
- from: "Billing Service"  
to: "taxes"
- from: "Content and Document Management Service"  
to: "Identity and Access Service"
- from: "Content and Document Management Service"  
to: "File Storage Service"
- from: "Commissioning Service"  
to: "Project Tracking Service"
- from: "Commissioning Service"  
to: "Workforce Management Service"

- from: "Commissioning Service"  
to: "File Storage Service"

- from: "Commissioning Service"  
to: "Notification Service"

- from: "Reporting Service"  
to: "Administration and Configuration Service"

- from: "Reporting Service"  
to: "Identity and Access Service"

- from: "Reporting Service"  
to: "Workforce Management Service"

- from: "Reporting Service"  
to: "Project Tracking Service"

- from: "Reporting Service"  
to: "CRM and Proposals Service"

- from: "Reporting Service"  
to: "Financial Management Service"

- from: "Reporting Service"  
to: "Billing Service"

- from: "Reporting Service"  
to: "Content and Document Management Service"

- from: "Reporting Service"  
to: "Commissioning Service"

- from: "Notification Service"  
to: "Administration and Configuration Service"

- from: "File Storage Service"  
to: "Administration and Configuration Service"